BUDT 758T: Data Mining and Predictive Analytics

Fall 2026, Tuesdays and Thursdays 2:00 to 3:15 PM, Van Munching Hall 1524
Instructor: Dr. Elena Marsh (fictional), office hours Wednesdays 1 to 3 PM

Grading

Homework (5 sets) 25%. Quizzes (3, in class) 15%. Midterm exam 20%. Team project 30% (proposal 5%, progress report 5%, presentation 5%, final report 15%). Final exam 10%.

Homework is submitted on ELMS by 11:59 PM on the date listed. Late homework loses 10% per day.

Schedule

Week	Date	Topic	Due
1	Tue 9/1	Course introduction, CRISP-DM	
1	Thu 9/3	Data preparation in pandas	
2	Tue 9/8	Exploratory analysis	Homework 1
2	Thu 9/10	Linear models	
3	Tue 9/15	Classification basics	Quiz 1 (in class)
3	Thu 9/17	Logistic regression	Homework 2
4	Tue 9/22	Decision trees	
4	Thu 9/24	Ensembles: bagging and boosting	Project proposal
5	Tue 9/29	Model evaluation	Homework 3
5	Thu 10/1	Cross-validation	Quiz 2 (in class)
6	Tue 10/6	Midterm review	
6	Thu 10/8	Midterm exam (in class, closed book)	Midterm exam
7	Tue 10/13	Clustering	
7	Thu 10/15	Association rules	Homework 4
8	Tue 10/20	Dimensionality reduction	
9	Thu 10/29	Text mining	Quiz 3 (in class)
10	Tue 11/3	Neural networks	Homework 5
12	Tue 11/17	Project work session	Project progress report
13	Thu 11/26	No class: Thanksgiving	
14	Tue 12/1	Team presentations	Project presentation

15	Thu 12/10	Course wrap-up	Final project report

Final exam: Tuesday, December 15, 1:30 to 3:30 PM, room to be announced.